

Table 2: Top Exposed Occupations According to GENOE

	1-year horizon	5-year horizon	10-year horizon
Telephone Operators	0.92	0.93	0.95
Data Entry Keyers	0.90	0.90	0.91
Telemarketers	0.87	0.88	0.89
Word Processors and Typists	0.86	0.87	0.90
Switchboard Operators, Including Answering Service	0.84	0.85	0.86
Credit Authorizers, Checkers, and Clerks	0.83	0.84	0.85
Meter Readers, Utilities	0.81	0.85	0.86
Motion Picture Projectionists	0.81	0.83	0.86
Order Clerks	0.80	0.82	0.84
Shoe Machine Operators and Tenders	0.77	0.84	0.84
Cutting and Slicing Machine Setters, Operators, and Tenders	0.80	0.82	0.83
Textile Winding, Twisting, and Drawing Out Machine Setters, Operators, and Tenders	0.78	0.82	0.84
Machine Feeders and Offbearers	0.75	0.81	0.84
Billing and Posting Clerks	0.76	0.80	0.84
File Clerks	0.72	0.82	0.85
Statistical Assistants	0.76	0.80	0.83
Travel Agents	0.76	0.80	0.82
Graders and Sorters, Agricultural Products	0.76	0.79	0.83
Office Machine Operators, Except Computer	0.72	0.81	0.83
Correspondence Clerks	0.76	0.78	0.82
Coil Winders, Tapers, and Finishers	0.74	0.78	0.83
Medical Transcriptionists	0.73	0.79	0.82
Bookkeeping, Accounting, and Auditing Clerks	0.72	0.79	0.83
Payroll and Timekeeping Clerks	0.74	0.77	0.82
Mail Clerks and Mail Machine Operators, Except Postal Service	0.73	0.78	0.82
Gambling Change Persons and Booth Cashiers	0.72	0.77	0.84
Tellers	0.70	0.79	0.83
Packaging and Filling Machine Operators and Tenders	0.70	0.79	0.83
Desktop Publishers	0.73	0.79	0.80
Grinding, Lapping, Polishing, and Buffing Machine Tool Setters, Operators, and Tenders, Metal and Plastic	0.74	0.76	0.81

Source: Authors' calculations using data from O*NET and surveys conducted with GPT-4 through OpenAI's API. *Note:* This table shows the occupations most exposed to AI, along with the GENOE estimates for the next one, five, and 10 years. Occupations were ranked based on the average exposure value across the three horizons.

Table 3: Less Exposed Occupations According to GENOE

	1-year horizon	5-year horizon	10-year horizon
Athletes and Sports Competitors	0.05	0.09	0.11
Oral and Maxillofacial Surgeons	0.04	0.11	0.16
Midwives	0.05	0.11	0.16
Firefighters	0.06	0.11	0.16
Kindergarten Teachers, Except Special Education	0.07	0.12	0.15
Special Education Teachers, Secondary School	0.09	0.11	0.18
Dancers	0.09	0.12	0.18
Clergy	0.10	0.12	0.17
Nurse Midwives	0.07	0.13	0.19
Elementary School Teachers, Except Special Education	0.08	0.13	0.18
Special Education Teachers, Middle School	0.09	0.14	0.18
First-Line Supervisors of Police and Detectives	0.09	0.14	0.18
First-Line Supervisors of Firefighting and Prevention Workers	0.09	0.13	0.19
Massage Therapists	0.09	0.13	0.18
Special Education Teachers, Preschool	0.09	0.14	0.19
Dentists, General	0.08	0.14	0.21
Pediatricians, General	0.09	0.15	0.20
Helpers–Roofers	0.09	0.16	0.20
Commercial Divers	0.09	0.17	0.19
Clinical Nurse Specialists & Advanced Practice Psychiatric Nurses & Acute Care Nurses & Critical Care Nurses & Registered Nurses	0.09	0.16	0.20
Child, Family, and School Social Workers	0.10	0.16	0.19
Education Administrators, Kindergarten through Secondary	0.10	0.16	0.19
Nursing Instructors and Teachers, Postsecondary	0.09	0.17	0.20
Roofers	0.10	0.17	0.20
Middle School Teachers, Except Special and Career/Technical Education	0.10	0.16	0.21
Judges, Magistrate Judges, and Magistrates	0.08	0.17	0.21
Preschool Teachers, Except Special Education	0.10	0.17	0.20
Structural Iron and Steel Workers	0.09	0.17	0.21
Hairdressers, Hairstylists, and Cosmetologists	0.09	0.17	0.22
Music Therapists & Art Therapists	0.11	0.17	0.20

Source: Authors' calculations using data from O*NET and surveys conducted with GPT-4 through OpenAI's API. *Note:* This table shows the occupations less exposed to AI, along with the GENOE estimates for the next one, five, and 10 years. Occupations were ranked based on the average exposure value across the three horizons.

Occupations in the 25th percentile, with relatively low AI exposure, include medical scientists, stonemasons, and occupational therapy aides, suggesting that roles requiring specialized knowledge or physical dexterity remain relatively insulated from AI replacement in the near term. The 75th percentile encompasses occupations with moderately high exposure, such as purchasing agents and tutors, indicating that even some knowledge-based professions face significant AI impact. Interestingly, farmworkers and criminal investigators represent the median level of AI

exposure, highlighting the broad reach of AI across both blue-collar and white-collar sectors.

Interestingly, our research reveals considerable heterogeneity in exposure within specific sectors, such as the healthcare industry. Pediatricians, neurologists, and general internal medicine physicians show low susceptibility to AI replacement, likely due to the high degree of human interaction, non-routine tasks, and complex decision-making required in these roles. In contrast, specialties like radiology, pharmacy, and dermatology face a higher potential for AI integration, which aligns with AI's growing capabilities in image analysis, data processing, and pattern recognition.

These findings underscore the importance of a granular, occupation-specific approach when assessing AI's potential impact on the workforce. They suggest that AI might reshape rather than wholly replace professions, with human expertise remaining crucial for complex cases and integrating specialized knowledge with broader contexts. This understanding is essential for effective workforce planning, educational curriculum development, and policy formulation, highlighting the need for targeted approaches to reskilling and upskilling initiatives that focus on cultivating distinctly human capabilities while leveraging AI's strengths.

Our analysis demonstrates high consistency and reliability in the GENOE estimates. As shown in Figure A.1 from the Appendix, the confidence intervals for the GENOE across the 100 iterations are remarkably narrow, indicating a high level of precision in the LLM's assessment of occupational exposure. Additionally, there is ordinal alignment across time horizons, with estimates for higher horizons being equal to or greater than those for lower horizons in almost all occupations. This pattern suggests a cumulative effect of AI exposure over time, with occupations generally becoming more susceptible to AI replacement as technology advances.

Importantly, our results show robustness to variations in the model's temperature setting. As observed in Table A.2 in the Appendix, the GENOE index remains unchanged whether it is calculated using iterations assigned higher temperatures or lower temperatures. This consistency across different levels of model creativity reinforces the reliability of our findings.

To address concerns about the "black box" nature of ChatGPT's assessments, we conducted follow-up surveys asking the model to explain its reasoning for each occupation's assessment in the five-year horizon. These responses allowed us to uncover patterns, ideas, and core topics influencing the models' decisions.

We identified at least five main patterns in LLM's rationale:

Technological Capabilities and Limitations. Many responses highlight what AI can currently achieve. For example, in the assessment of "File Clerks," it notes AI's current proficiency in data entry and document management, leading to a high likelihood of replacement (0.9). On the other hand, future advancements are often considered, especially regarding the merging of robotics and AI, a topic mentioned in the surveys 236 times—not only to see the potential, but also to cast doubts about the feasibility of AI replacement. For instance, in the case of "Barbers," ChatGPT notes that while advances in AI and robotics have made strides in automating various tasks, the signif-

icant manual dexterity, precision, and personal interaction involved in barbering are challenging to replicate with current or medium-term advances in technology.

Tasks Nature (Routine vs Non-Routine). The model consistently distinguishes between routine and non-routine tasks. Occupations with routine, repetitive tasks are considered more likely to be replaced by AI. The term "routine" appears 72 times in the responses, indicating a strong emphasis on the predictability and repetitiveness of tasks. For example, the response for "Anesthesiologists" (likelihood 0.2) notes that although AI might assist with some routine tasks such as diagnostic procedures, monitoring patients, and scheduling, the holistic and critical (non-routine) nature of anesthesiology, including patient care, emergency response, coordination with medical teams, and individualized decision-making, presents substantial challenges to full automation.

Ethical Considerations. The model frequently raises ethical concerns, particularly for occupations involving direct human interaction or safety-critical roles. The terms "ethical" and "safety" appear 28 and 81 times, respectively, in the responses. For example, in the case of "Judges," ChatGPT notes that the complex and highly ethical nature of judicial tasks, such as interpreting laws and making nuanced judgments, poses significant challenges for AI, making it unlikely that judges will be replaced by AI soon (likelihood of 0.2). Similarly, in occupations like Pilots and Amusement and Recreation Attendants, safety concerns are raised to lower AI prospects of replacement.

Regulatory Environment. Legal and regulatory factors are also considered. Regulatory concerns were raised 63 times in the follow-up responses. This occurs in occupations like bakers, drivers, and civil engineers. Heavily regulated industries might be less likely due to strict regulations and standards that AI must meet.

Cost-Benefit Analysis. Economic feasibility also plays a role. For instance, jobs that require high (low) levels of human expertise and are less (more) economically viable for automation may have lower (higher) likelihoods. This factor is considered in occupations such as "Farm Equipment Mechanics and Service Technicians" and "Hand Packers and Packagers."

These findings not only provide insight into the model's decision-making process but also align with established theories in labor economics and technological change. The consideration of task routineness, for instance, echoes the task-based approach to technological change (Autor *et al.*, 2003, 2006; Acemoglu and Autor, 2011; Autor, 2013), while the emphasis on ethical and regulatory factors reflects the growing awareness of AI's societal implications.

3.2 Comparison with Existing Indexes

Figure 2 compares the GENOE index with other indexes of computerization and occupational exposure to technological displacement.⁶ Interestingly, our index shows little correlation with

⁶Since most indexes were calculated using the SOC-2010 classification, we transformed the GENOE from SOC-2018 to SOC-2010 using the crosswalks from the Bureau of Labor Statistics for comparison purposes. Henceforth, in this document, whenever the generated indexes are compared to existing indexes, they are transformed to SOC-2010.